



Data processing & Data pipelines

Session 4

FLOWMINDER.ORG

NOMMON



Overview

This session covers the following topics:

1. Stages of a mobile phone data pipeline
2. Inputs to the Data Pipeline
3. Privacy-by-design in mobile phone data pipelines
4. Structures of mobile phone data pipelines

Learning objectives

- Identify the stages of a data pipeline for mobile phone data
- Identify the key data inputs and where they enter the data pipeline
- Classify the sensitivity of data in a mobile phone data initiative at different levels of processing
- Identify the stages of the data pipeline at which personal and commercially-sensitive data are being processed
- Explain which stakeholders are responsible for different stages of processing mobile phone data depending on the data access model agreement and approach to implementing the data pipeline
- Describe the key considerations for different implementations of the data pipeline, based on data access model choices made, in particular those related to data privacy and security

Stages of a Mobile Phone Data Pipeline

From Network Events to Statistics

Collection

Networks events (e.g. calls, sms, data sessions, location updates etc) occur on MNOs network

MNO records these events for operational purposes (e.g. billing, network monitoring)

ETL

MNO pseudonymises the data

Agreed data features are extracted and transformed into the necessary format

Quality assurance checks (syntactic and semantic)

Processing

Data cleaning and filtering

Data sampling

Continuity models

Detecting meaningful locations (home/work etc)

Routing

Aggregation

Spatial and temporal aggregation based on domain specific methodology

Bias Adjustment and Scaling

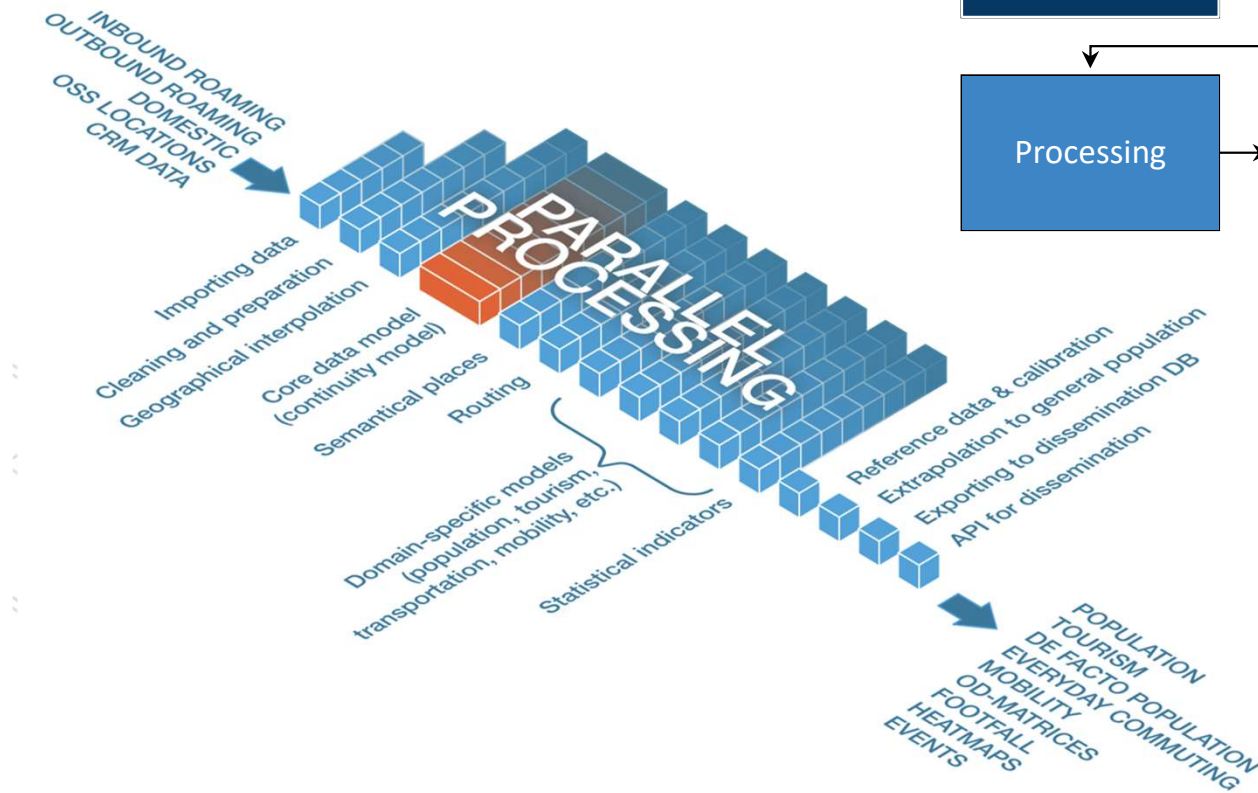
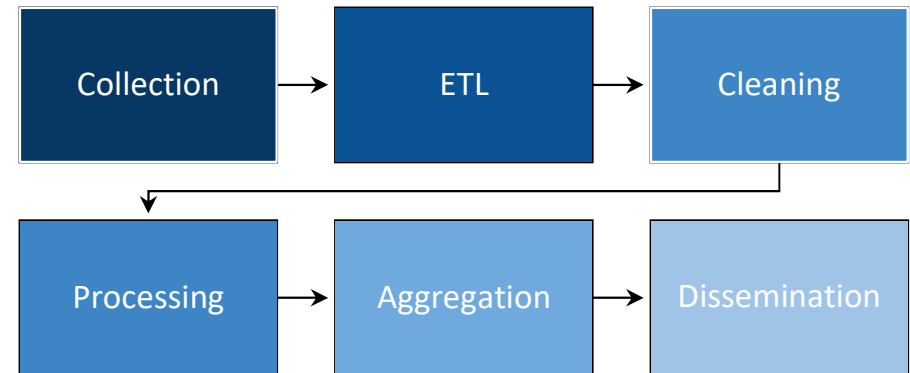
Applied statistical methods to account for biases in the data

Scale estimates from subscribers to population

Quality assurance

Statistical disclosure controls

Processing steps

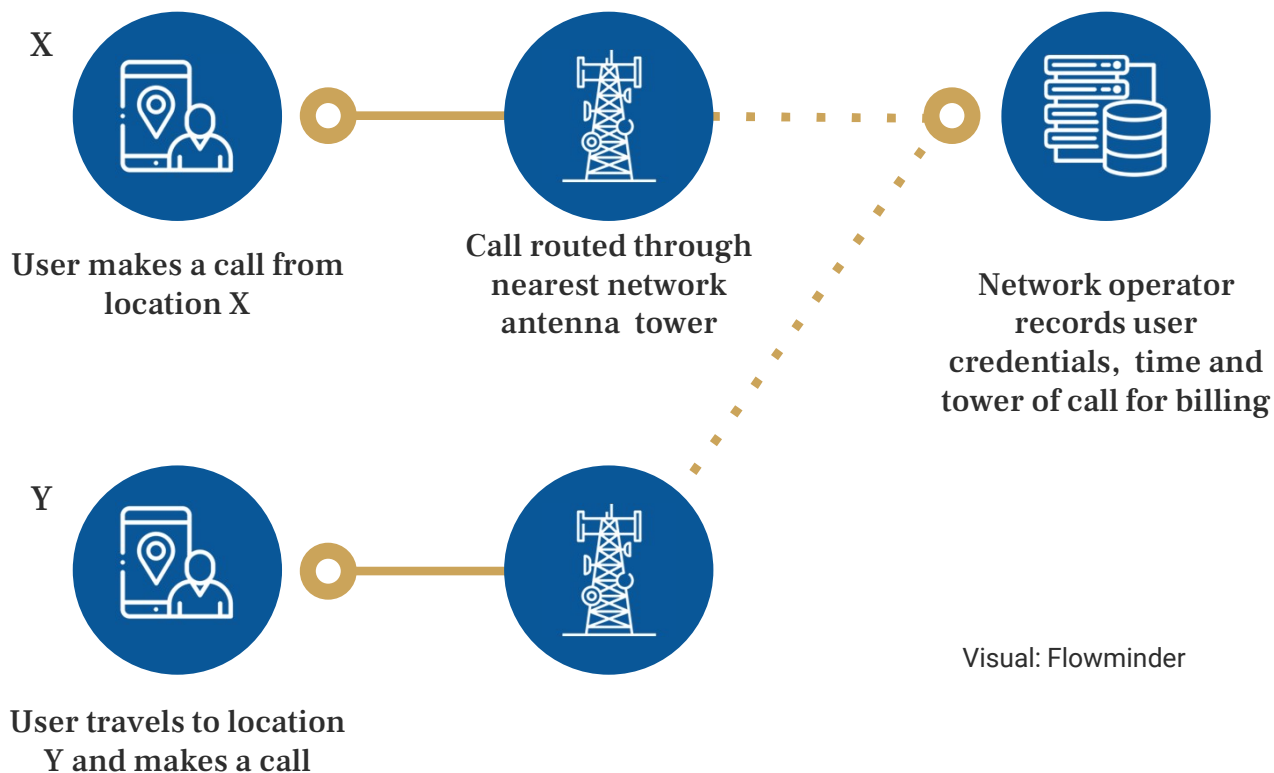


Collection (CDR example)

Collection starts automatically as mobile phones are used

When we turn on our phones this information is registered in the network so when someone calls us, the network knows where to route the call

Here is an example of how two events are logged at a mobile network operator (MNO)



Each mobile phone
generates many events
during the day and mobile
network stores billions of
events

- Subscriber identity
- Time
- Antenna or Cell ID

Subscriber_D		Time	Cell_D
Collection	8347592345023760	2025-09-24 09:31:12	14614
	8347592345023760	2025-09-24 09:45:07	14614
	8347592345023760	2025-09-24 10:00:25	14880
	8347592345023760	2025-09-24 10:04:51	14920
	8347592345023760	2025-09-24 10:08:27	14711
	8347592345023760	2025-09-24 10:10:31	14760
	8347592345023760	2025-09-24 10:13:34	14084
	8347592345023760	2025-09-24 10:15:27	14715
	8347592345023760	2025-09-24 10:18:53	14486
	8347592345023760	2025-09-24 11:08:00	14486
	8347592345023760	2025-09-24 11:20:31	14486
	8347592345023760	2025-09-24 12:16:06	14486
	8347592345023760	2025-09-24 14:08:35	14486
	8347592345023760	2025-09-24 14:45:35	14486
	8347592345023760	2025-09-24 15:21:46	14486
	8347592345023760	2025-09-24 15:21:46	14486

 Cell tower coverage Cell towers

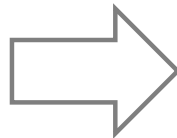
ETL: Preparation

```
<?xml version="1.0"?>
<cdr>
  <id>b478fndh4dcvjt</id>
  <start_time>2014-04-23 14:34:12</start_time>
  <start_epoch>1217529338</start_epoch>
  <start_uepoch>1217529338212616</start_uepoch>
  <answer_epoch>1217529338</answer_epoch>
  <answer_uepoch>1217529338452698</answer_uepoch>
  <end_epoch>1217529377</end_epoch>
  <end_uepoch>1217529377795951</end_uepoch>
  <end_time>2014-04-23 14:35:14</end_time>
  <billsec>62</billsec>
  <success>1</success>
  <imsi>384845321548865</imsi>
  <type>call_out</type>
  <type_voice</type_voice>
  <gen>2g</gen>
  <ogt>248487865145</ogt>
  <msisdn_rec>284872535485</msisdn_rec>
  <default_area_code>24807</default_area_code>
  <duration>00:01:02</duration>
  <record_type>125425</record_type>
  <accountcode>1460</accountcode>
  <user_context>default</user_context>
</cdr>
```

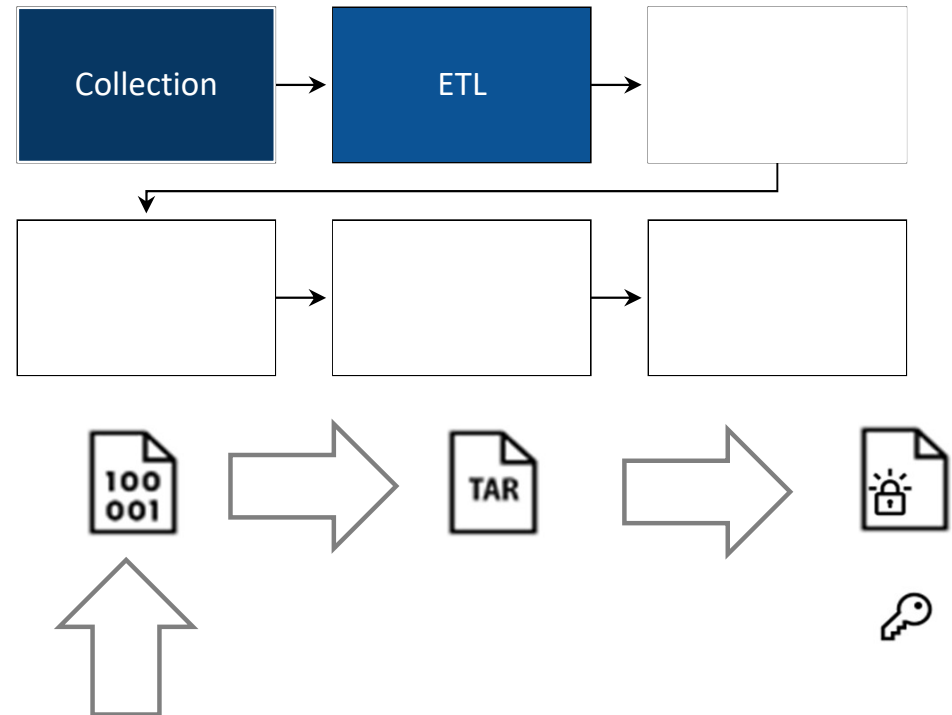


Pseudonymisation

```
id;time;cell_id;iso_a2
5542330018;2013-01-01 14:31:12;KCPBI14;FI
5542330018;2013-01-01 14:31:12;KCPBI14;FI
7775564918;2013-01-01 14:37:33;KCPBI14;SE
9909558001;2013-01-01 16:02:03;LDUCA14;RU
3812036960;2013-01-01 18:26:03;TELBI13;DE
1979419073;2013-01-02 01:01:06;LDUCA14;GB
3677654334;2013-01-02 06:06:50;ABBDA24;LV
3677654334;2013-01-02 06:06:50;ABBDA24;LV
5582179296;2013-01-02 12:29:12;TELBI13;US
7683234766;2013-01-02 15:58:58;KCPBI14;AU
7437275049;2013-01-02 17:07:00;PHTGR20;AT
8246667432;2013-01-02 17:46:39;HHOCTX0;DK
8246667432;2013-01-02 17:46:39;HHOCTX0;DK
```



```
12480017593759052014091608481224800116224212
68465334888220116735205390459361466302902486
88611241518507953473827391657145825956967686
61498550428263044677386749601140377218542630
83641584575223884289417071576240118494724902
17805293905217303576952739336266040996358810
79906289886398399954774325251381308505266619
26804156077180956420453516287816126347213625
```

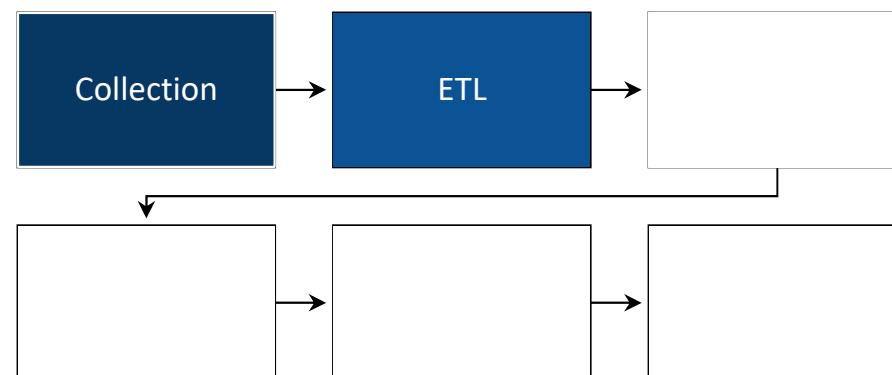


ETL: Preparation steps

Before transferring data it needs to be prepared

- Formatting the data into **standard agreed format**
- Hashing IMSI (or whatever other ID-base) - **Pseudonymisation**
 - Same pseudonymised identifiers are consistently applied across the dataset to allow for longitudinal analysis.
- Extracting MCC from IMSI (Inbound data)
- **QA:** Checking data integrity, missing data
- Minimise (binary, pack)
- **Encrypt**

ETL: Transmission

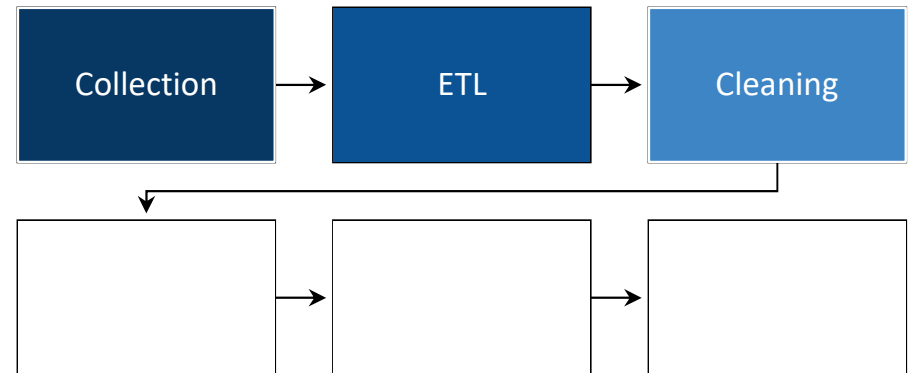


ETL: Transmission security

- Encryption: Data must be **encrypted** with strong encryption during transmission stage
- Data Volume: Volumes are huge so consideration of data size and capacity for efficient transmission must be taken into account (e.g. partitioned batch)
- Authorization: Use of strong identity verification and access control mechanisms between two organizations
- Data Transfer Protocol: **Secure file transfer protocols** (e.g., SFTP, HTTPS) with possible IP restrictions ensure encrypted and safe transmission of data
- Data Integrity and Validation: Use checksums or hash values to **validate data integrity** during transmission

Cleaning

In pre-processing stage all possible data errors and non human activity is removed



Why should data be cleaned?

- Wrong data can give way to wrong decisions, conclusions and poor analysis.
- Data cleaning can remove many errors and inconsistencies with data.
- Reducing the amount of data is also beneficial for productivity.

Example filters for cleaning the data:

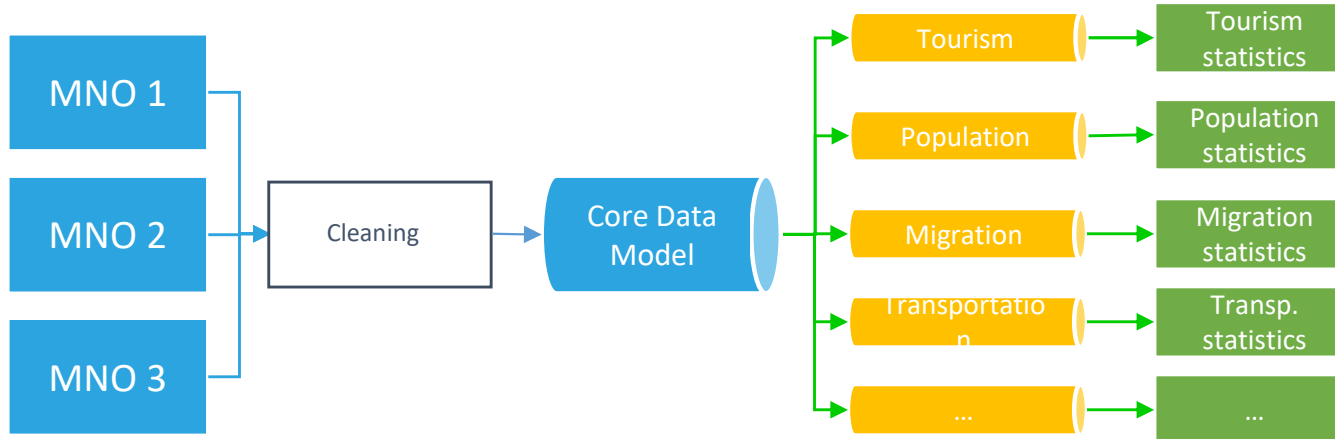
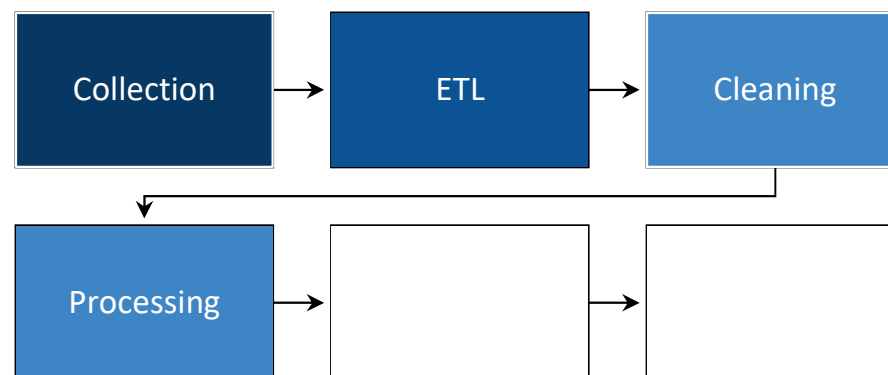
- IoT
- Maritime
- Speed
- Ghost Events
- Accidental Roaming

Cleaning filter examples

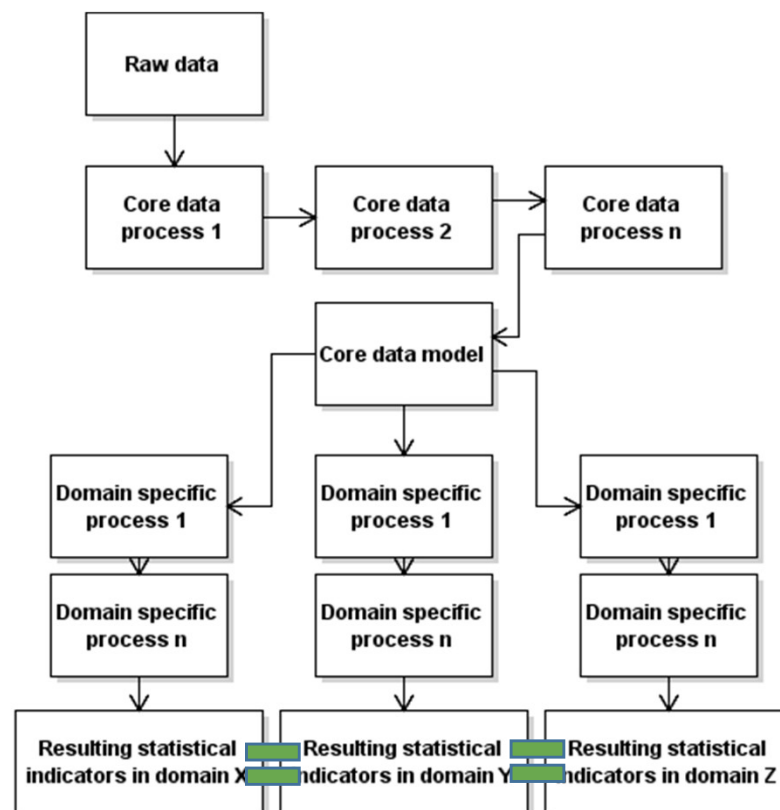
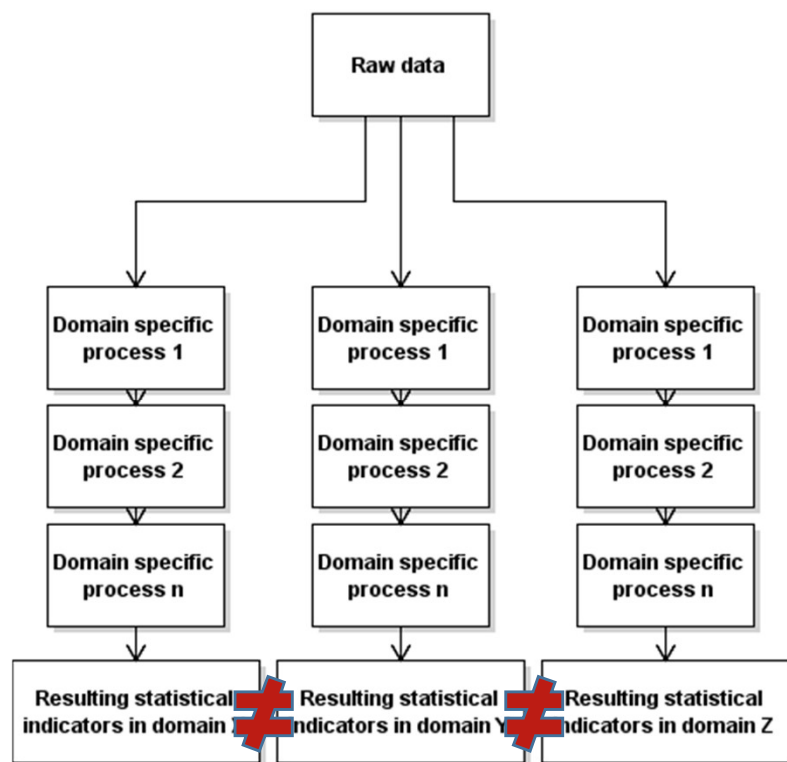
- **IoT** - Eliminate data from IoT devices (e.g. home security systems)
- **Maritime** - Account for individuals at sea connecting to land antennas to avoid miscounting tourists
- **Planes** - Exclude data generated by phones connecting to ground masts during flights
- **Ghost Events** - Remove erroneous events generated by the MNO system
- **Accidental Roaming** - Isolate instances where users accidentally connect to a foreign network near borders, as this is not tourism
- ...

Processing

- From raw data to domain indicators
- Create data model to represent reality as realistically as possible, but taking into account the data limitations



Processing: Siloed or Centralized (core model)



Without Continuity Model

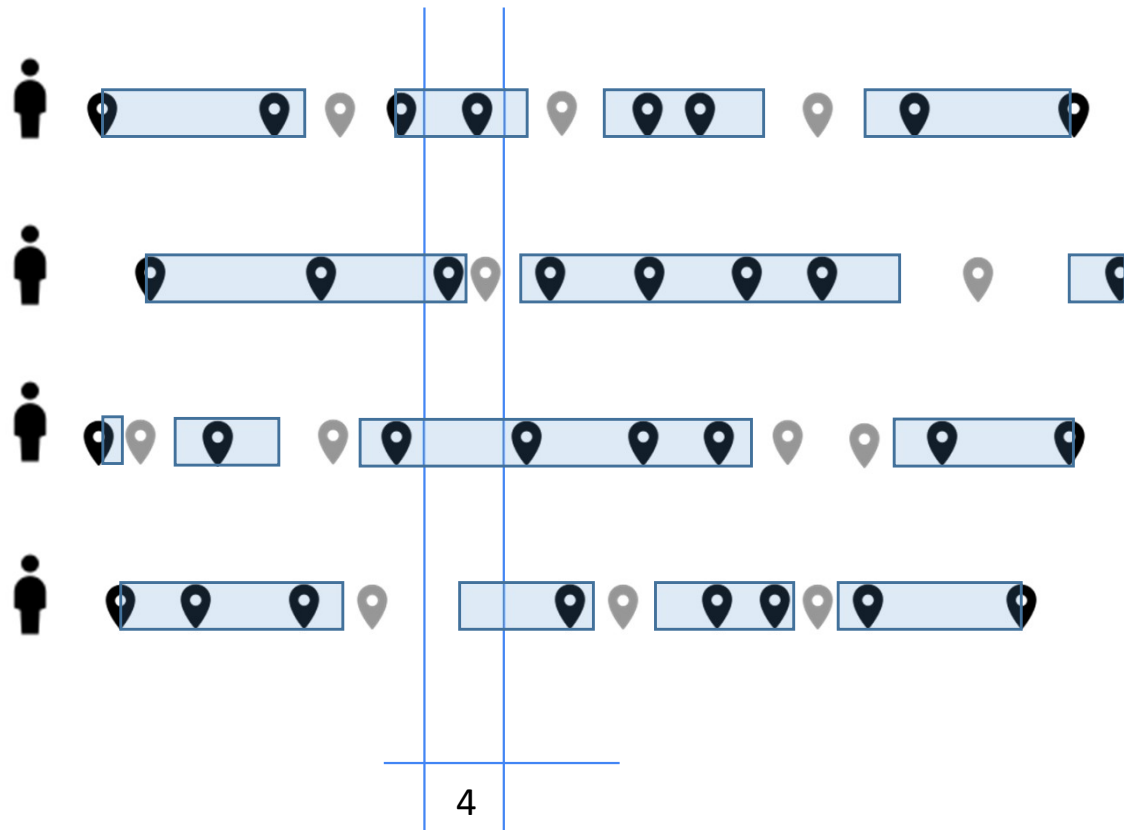


📍 Location A

📍 Other location

With Continuity Model

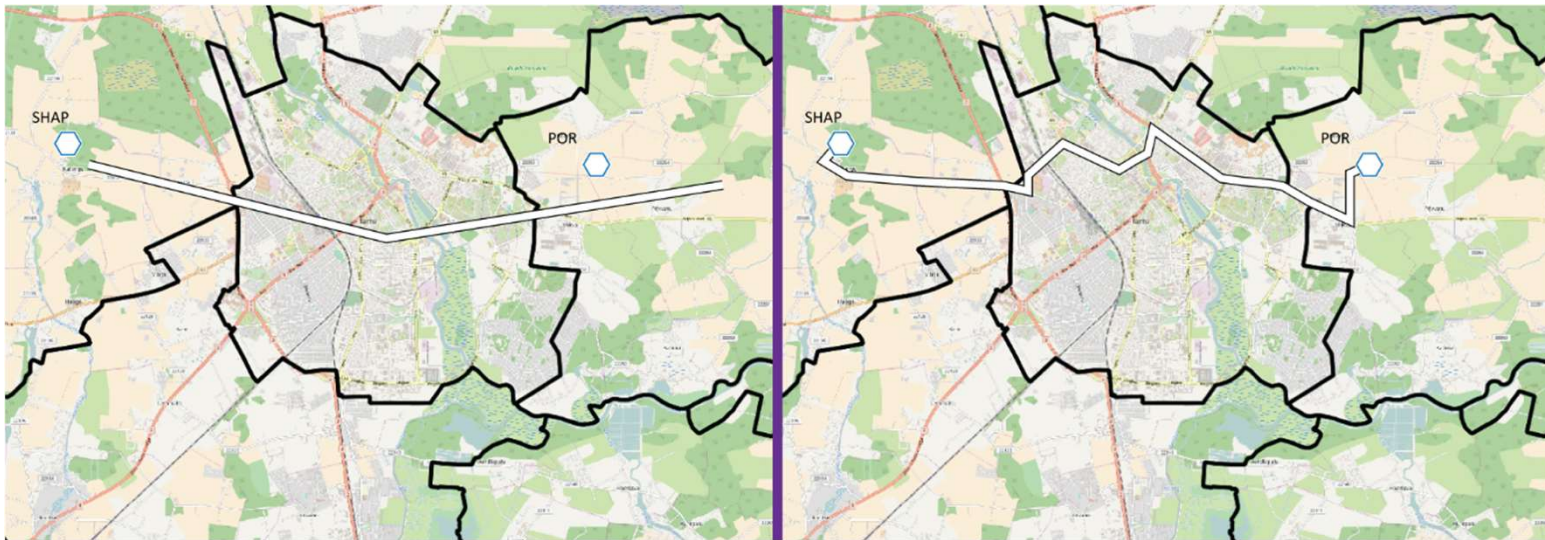
 Location A
 Other location



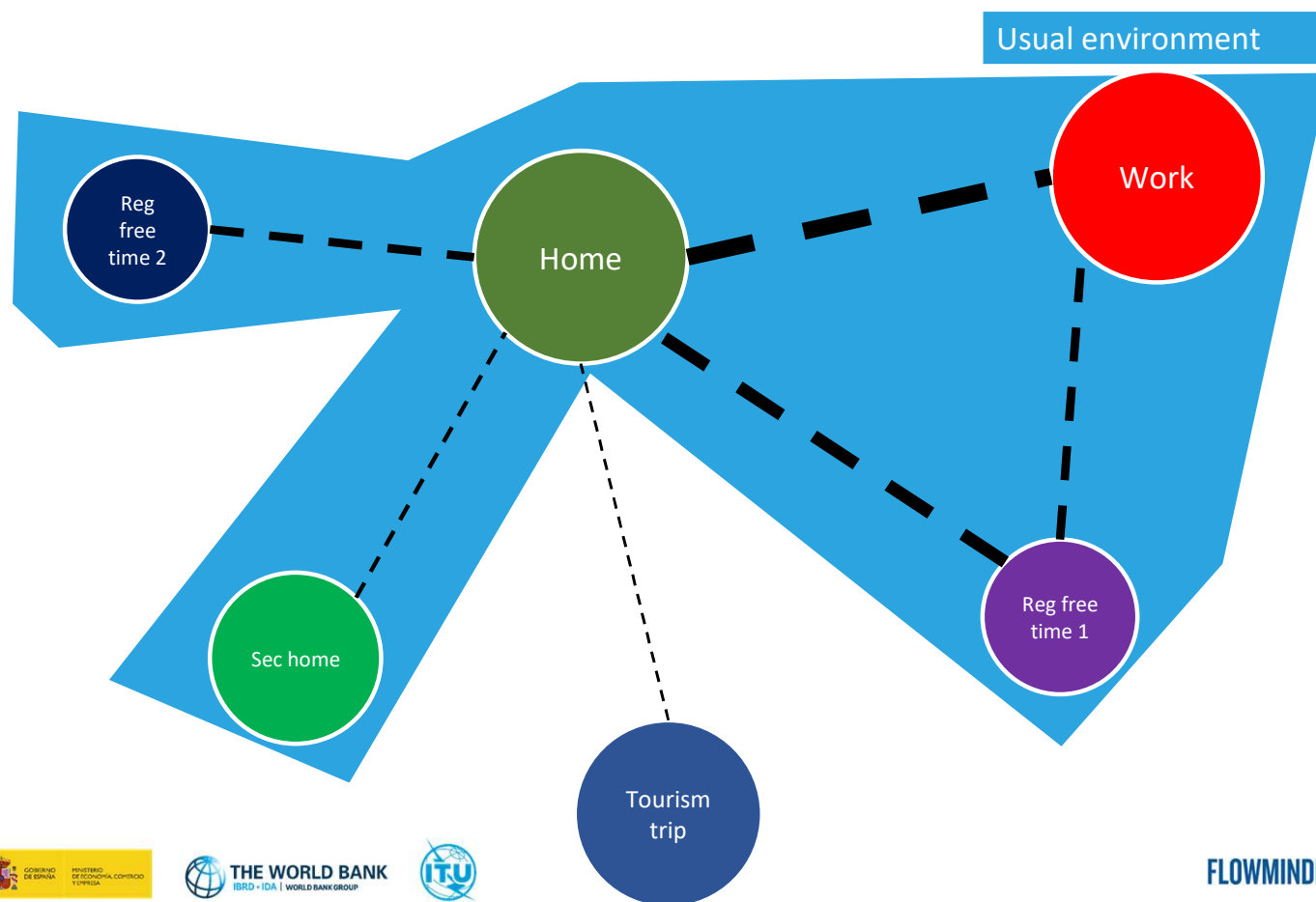
Stay and Move sections

In essence, the spatio-temporal behaviour can be broken down to stay and move sections

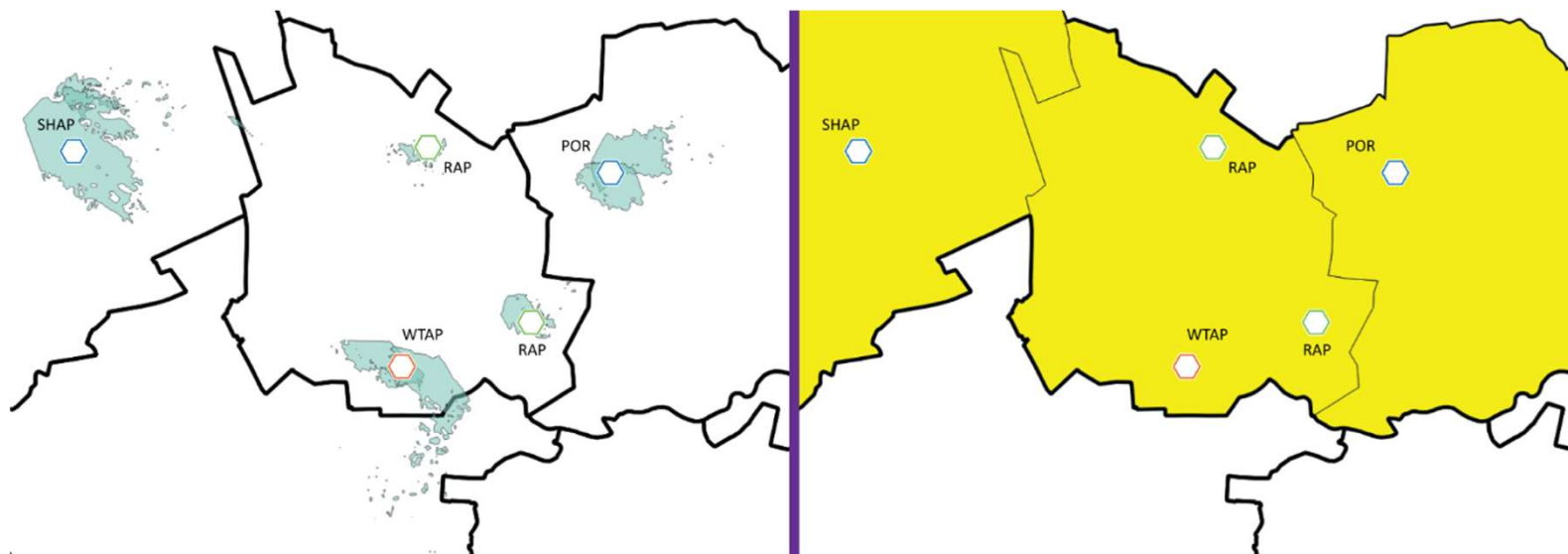
For movement sections sometimes we need to identify the (probable) road



Identification of Meaningful Locations (Anchors)



Identification of the Usual Environment



Interpolation of locations

Improving location data through probabilistic model using additional data layers:

- Buildings and land use classes for stays
- Roads for move sections

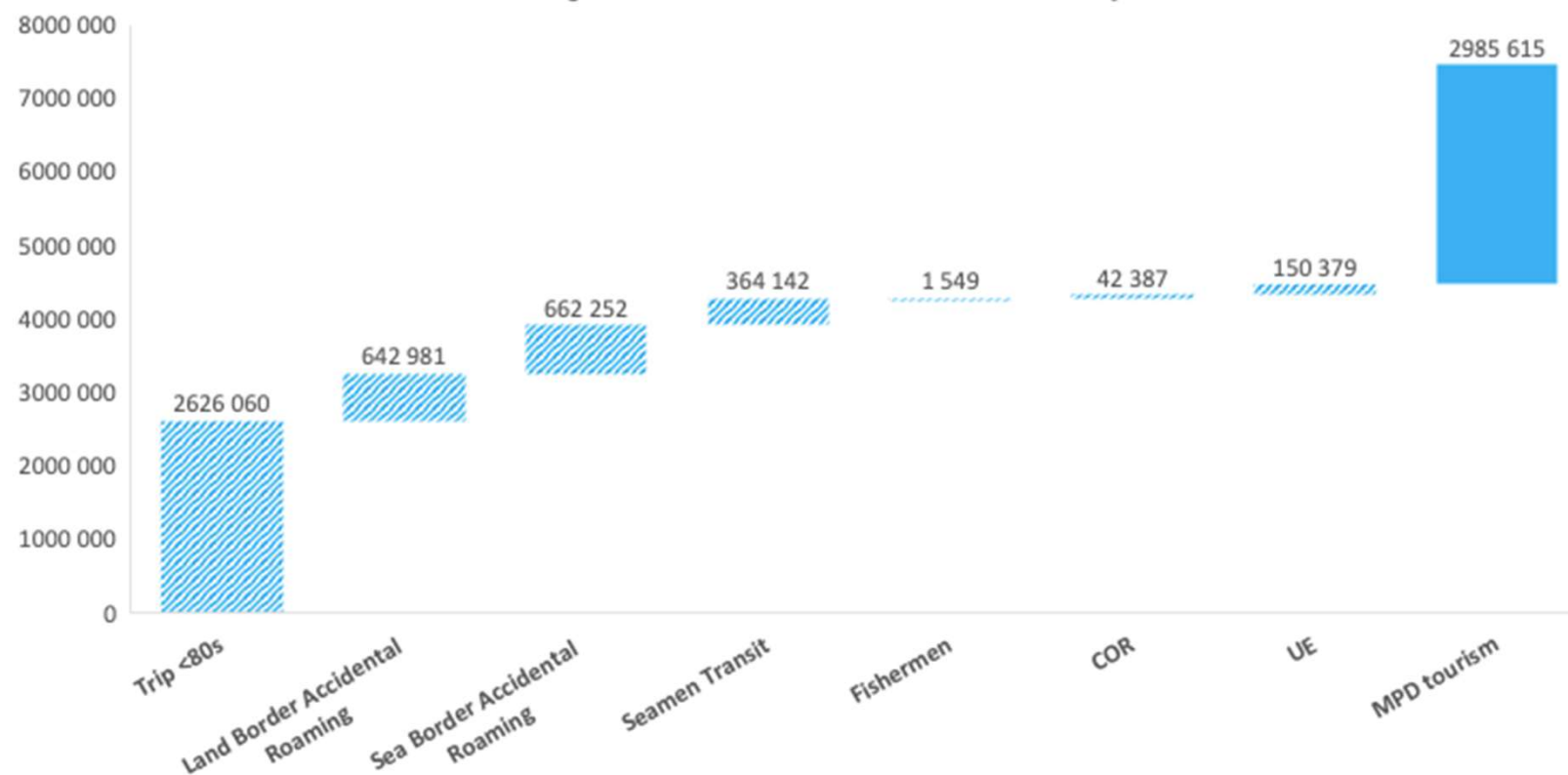


Other processing step examples

- Functional midnight
- “Take-home” action (CDRs)
- Identification of entry and exit gates
- Identification of tourism trips
- Identification of domestic trips
- Routing move sections
- Distinguishing between moving ‘stops’, transit stops and actual stops
- Long-term migration
- Cross-border usual environment
- Functional classifications in locations
- Modal split
- ...

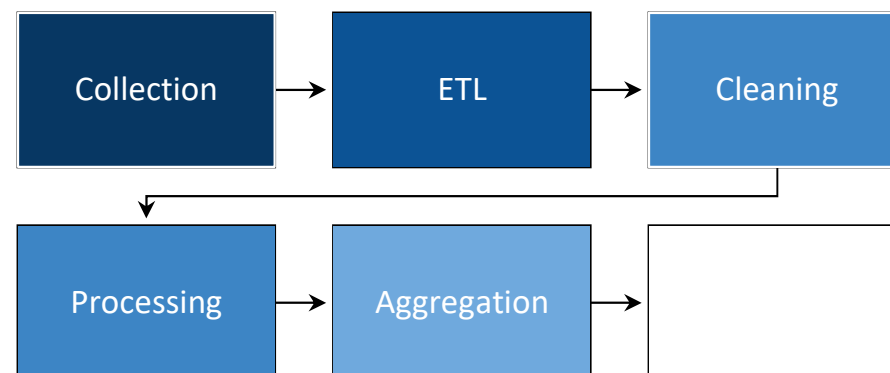
Filtering of non-tourism activity from MPD

Cascading of MPD data across error classes, one year



Aggregation

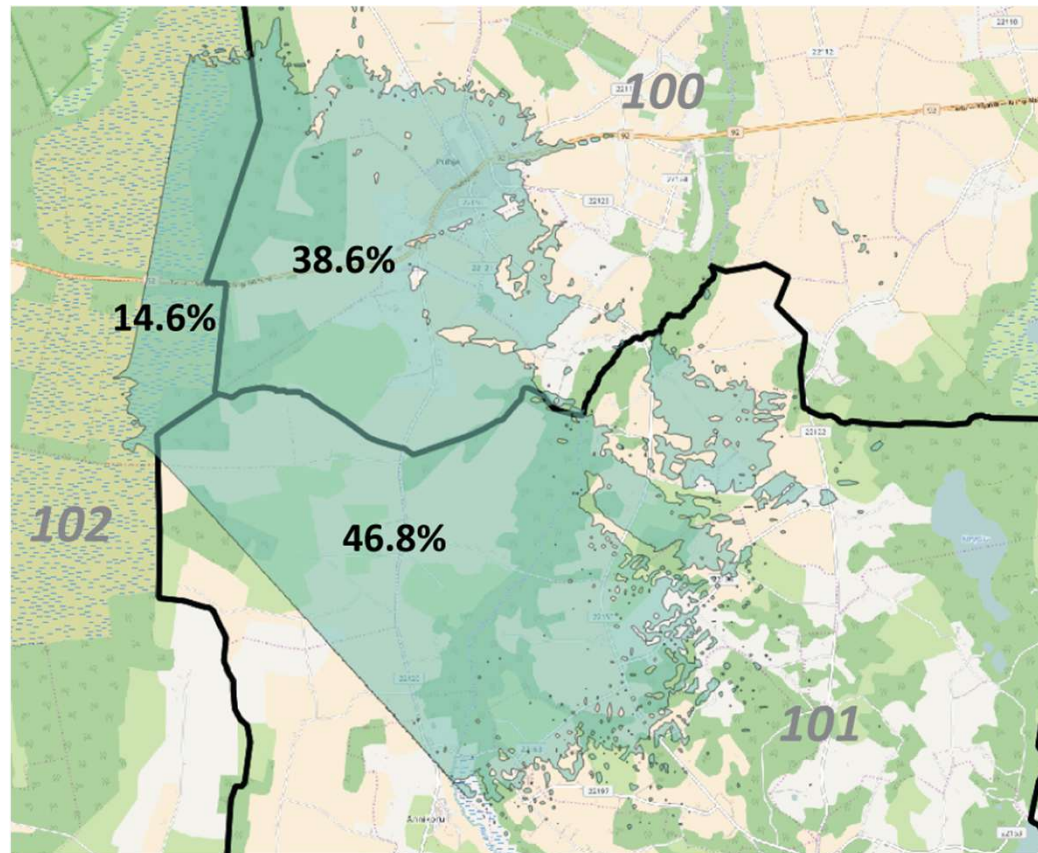
- Spatial
 - LAU0 (Country)
 - LAU1
 - LAU2
 - LAU3
 - Grid
- Temporal
 - Year
 - Quarter
 - Month
 - Week
 - Day
 - Hour
- Country of Residence



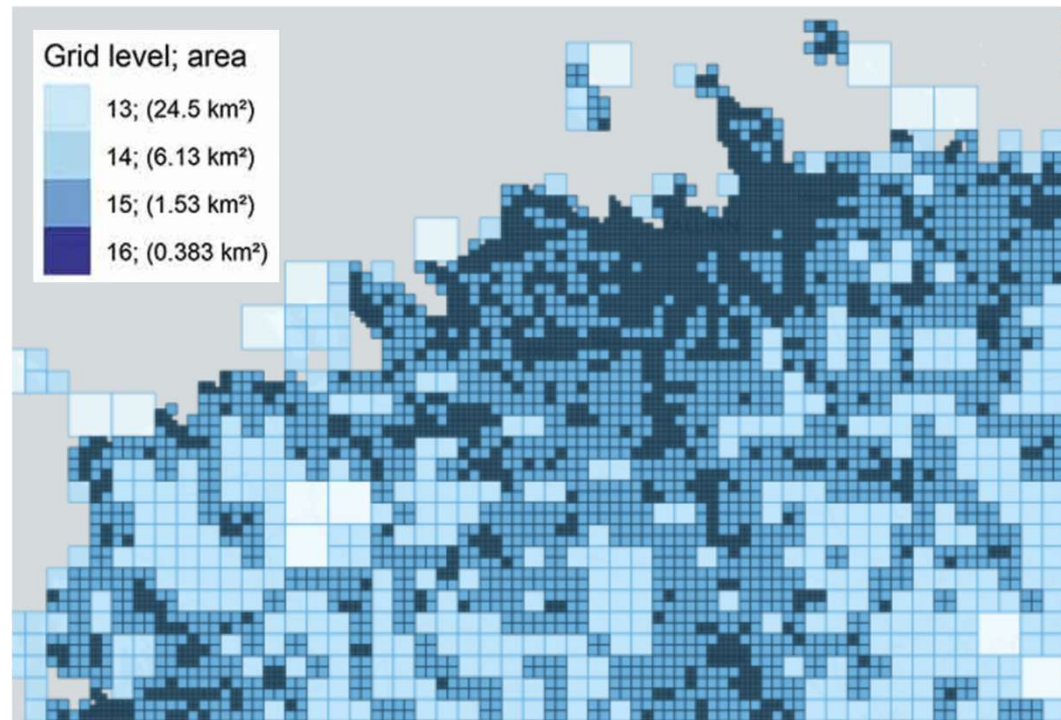
Tourism statistics aggregation examples:

- | | |
|--------------------|-----------------|
| ● Inbound/Domestic | ● Trips started |
| ● Visitors present | ● Trips present |
| ● Unique visitors | ● Trips ended |
| ● Visits started | ● Nights Spent |
| ● Visits present | ● Days Spent |
| ● Visits ended | ● ... |

Aggregation. Spatial: Local Administrative Unit (LAU)



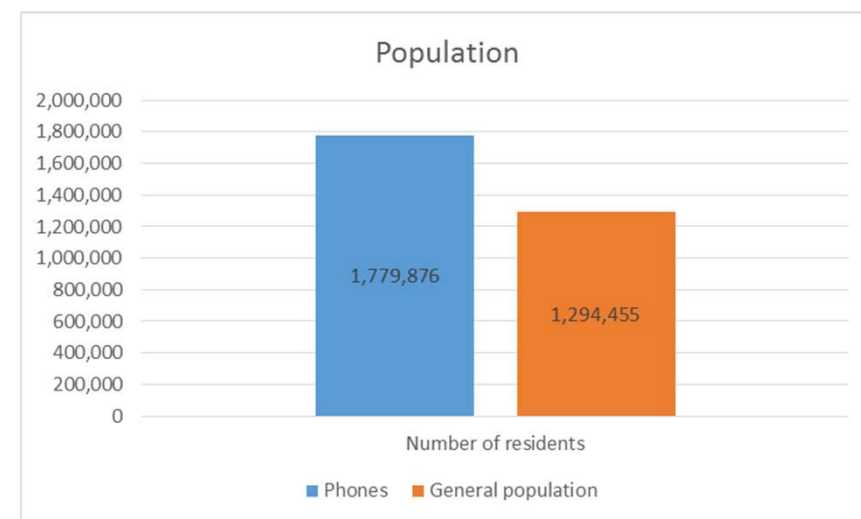
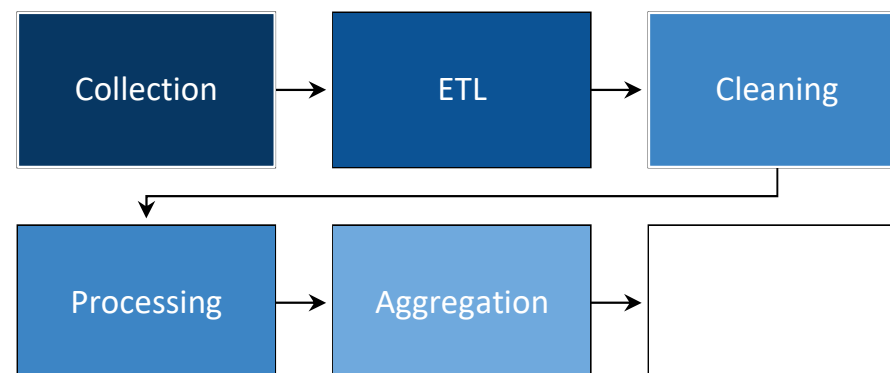
Aggregation. Spatial: Grid



Adaptive grid example from Tallinn Metropolitan Area from a research paper
“Spatial interpolation of mobile positioning data for population statistics”
<https://doi.org/10.1080/17489725.2021.1917710>

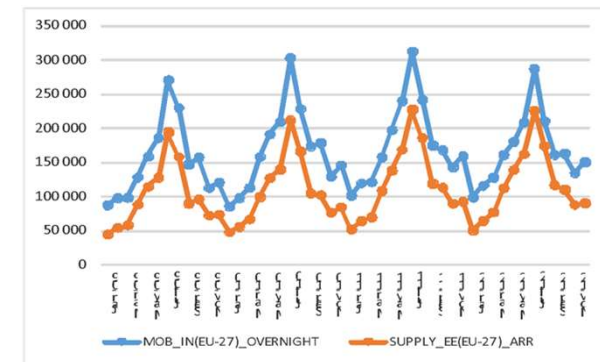
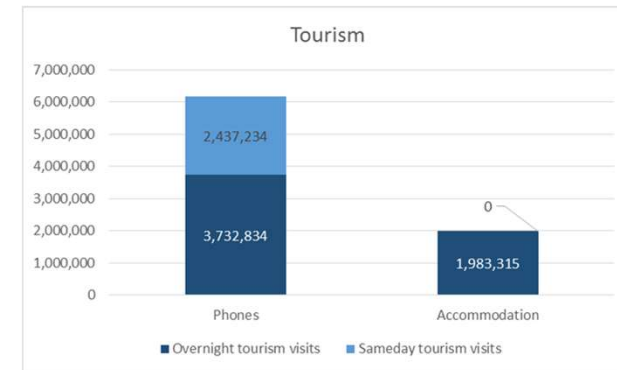
Population-Scaling and Bias Adjustment

- Market shares of the MNOs
- Penetration of SIM cards in the country / regions
- Reference data (baseline, ground truth, surveys)
- Calibration at specific points in time (aggregates vs reference data)
- Estimate using calibration coefficients matrix



Bias Adjustment and Scaling

- Statistical techniques are used to adjust for selection biases in the data
- Estimates derived from subscriber data are scaled to represent the entire population.
- Validation and plausibility checks (QA) are performed to ensure the accuracy, completeness, and reliability of the aggregated results.
- Data protection measures (SDC) are applied to prevent the identification of individuals in the published outputs



Computing scaling factor - The Gambia example

Scaling factor (SF) for the administrative unit (District i):

$$\text{SF}_i = \text{Resident}_{\{i, \text{allpop-census}\}} / \text{Resident}_{\{i, \text{subscribers-census}\}}$$

where

$\text{Resident}_{\{i, \text{allpop-census}\}}$: Total residential population of District i [Census]

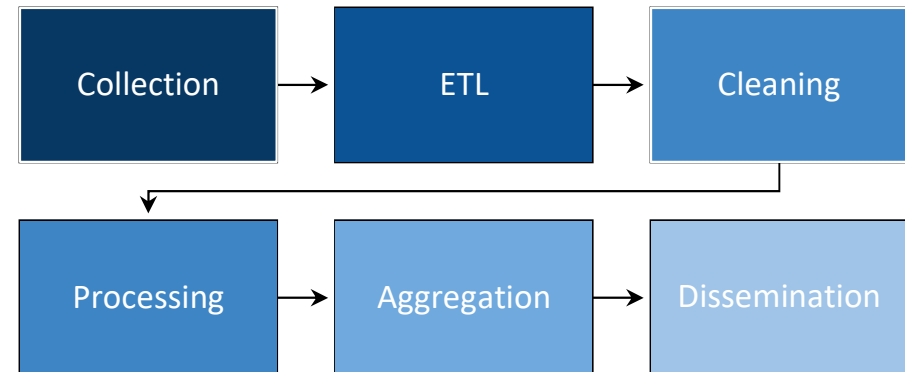
$\text{Resident}_{\{i, \text{subscribers-census}\}}$: Number of subscribers residing in District i [Census]

$\text{Resident}_{\{i, \text{subscribers-CDR}\}}$: Number of subscriber estimated to reside in District i [CDR]

$$\begin{aligned} &\text{Adjusted residential population estimate} \\ &= \text{SF}_i \times \text{Resident}_{\{i, \text{subscribers-CDR}\}} \end{aligned}$$

Dissemination

- Reports
- Tables
- Maps
- Dashboards
- API-s



month	country	d_class	unique_visitors	stays_started	spent_nights	spent_days
01.09.2017	FI	1	6366	9447	0	9447
01.09.2017	FI	2	30735	42849	167672	205225
01.09.2017	FR	1	904	1001	0	1001
01.09.2017	FR	2	5704	8672	31783	39465
01.09.2017	GB	1	635	692	0	692
01.09.2017	GB	2	4355	5297	25144	29866
01.09.2017	GE	1	227	242	0	242
01.09.2017	GE	2	1230	1330	7160	8105
01.09.2017	IE	1	82	89	0	89
01.09.2017	IE	2	464	564	2708	3244
01.09.2017	LT	1	1185	1325	0	1325
01.09.2017	LT	2	7163	8789	16769	23788
01.09.2017	LU	1	74	102	0	102
01.09.2017	LU	2	316	414	1764	2145
01.09.2017	LV	1	8266	10083	0	10083
01.09.2017	LV	2	20518	24894	42843	62104

Eesti Pank
Statistical indicators
Estonian Bank > Statistical indicators > External sector statistics > International travel statistics

Table Figure

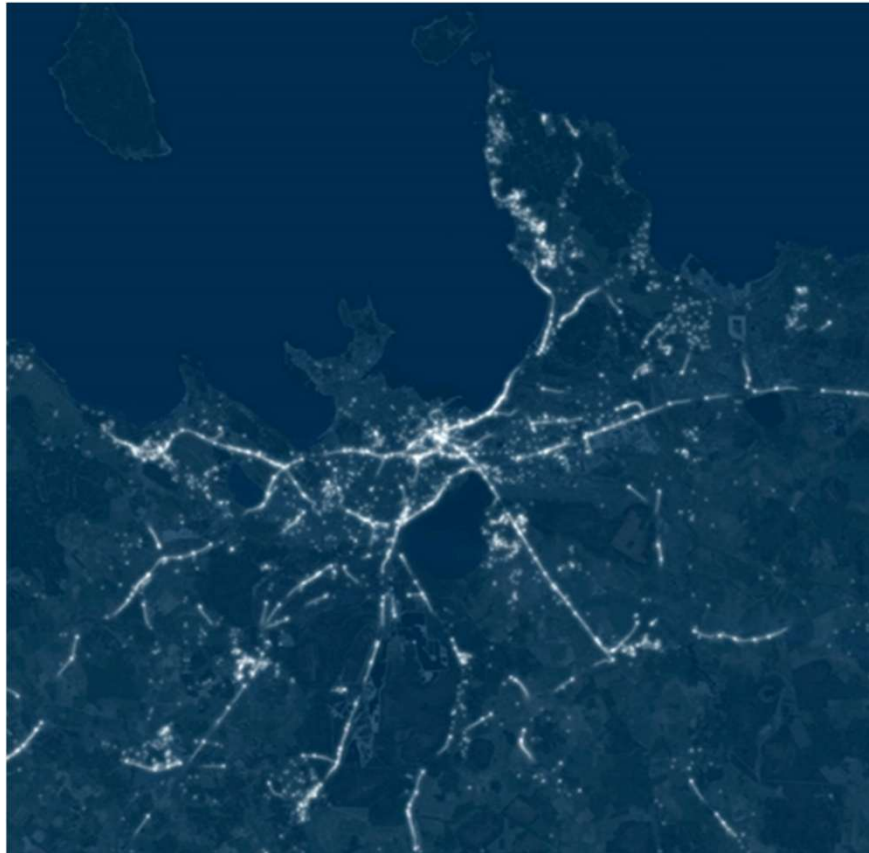
COUNTRY: display all INDICATOR: display all STEP: quarter FROM: 2024 TO: 2024

DISPLAY DOWNLOAD SEND

1. Inbound travel updated 08/08/2024

	Q1/2024				Q2/2024			
	Total number of visits	Number of same-day visits	Number of overnight visits	Length of overnight visits (nights)	Total number of visits	Number of same-day visits	Number of overnight visits	Length of overnight visits (nights)
TOTAL	722,525	270,545	451,980	1,590,215	1,142,798	449,899	692,899	2,322,824
EU 27	612,800	233,824	378,976	1,176,099	939,195	373,520	565,675	1,774,154
CIS	31,017	9,497	21,520	181,710	35,258	10,809	24,449	158,687
AU Australia	2,762	1,667	1,095	6,426	5,835	3,128	2,707	10,814
AT Austria	3,047	895	2,152	9,142	6,454	1,669	4,785	17,520
BE Belgium	3,780	1,125	2,655	10,111	7,406	1,824	5,582	20,423
CA Canada	1,552	525	997	3,460	5,191	2,789	2,402	7,701
CN China	4,697	2,517	2,180	8,047	8,645	4,124	4,521	12,668
DK Denmark	3,824	982	2,842	12,577	8,306	1,527	6,779	25,876
FI Finland	318,075	139,447	178,628	448,117	491,972	219,573	272,399	737,774
FR France	9,617	3,311	6,306	34,315	17,594	4,139	13,455	65,932
DE Germany	25,328	7,368	17,960	78,810	73,414	27,187	46,227	174,301
IT Italy	10,645	3,355	7,290	27,866	17,125	5,377	11,748	40,036
JP Japan	3,025	1,469	1,556	4,921	4,806	1,842	2,964	11,502
KR Korea, Republic of	1,072	755	317	1,441	4,095	2,353	1,742	3,348
LV Latvia	136,478	42,326	94,152	273,238	145,885	58,612	87,273	297,321

Dissemination animation example



Inputs to the Data Pipeline

The diagram illustrates the location area concept in GSM. A Mobile Device (MD) is shown moving through a Location Area (LA) defined by several BTS/NodeB (antenna/cell) coverage areas. The diagram shows the MD's path and the coverage areas of the cells it traverses. Below the LA, two BSC/RNC/MME blocks are shown, connected to the cells. A note indicates that other BSCs, RNCs, and MMEs of the same MNO exist.

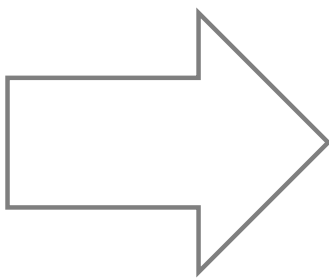
The diagram illustrates the Network Architecture of the EPC (Evolved Packet Core). It shows various network elements and their interconnections:

- SGSN** (Service GPRS Support Node) for internet traffic (2G, 3G)
- SGW** (Serving Gateway) for internet traffic (4G)
- VLR** (Visitor Location Register / MD BSC/RNC/MME last location)
- HSS (4G)** (Home Subscriber Server / MD LA last location)
- HLR** (Home Location Register / MD LA location)
- AuC (2G, 3G)** (Authentication Centre / authentication of subscriber)

Connections are shown between these elements, indicating the flow of information and services.

Figure 1 illustrates the network architecture of a mobile network. The diagram shows the following components and their connections:

- IP Network (internet):** Represented by a cloud icon, it is connected to MNOs and partner MNOs.
- MNOs (Mobile Network Operators):** Represented by server icons, they are connected to the IP Network and the PSTN.
- partner MNOs:** Represented by server icons, they are connected to the IP Network and the PSTN.
- PSTN (Public Switched Telephone Network):** Represented by a box icon, it is connected to MNOs and partner MNOs.
- BSC (Base Station Controller):** Represented by a box icon, it is connected to the PSTN and the RNC.
- RNC (Radio Network Controller):** Represented by a box icon, it is connected to the BSC and the MME.
- MME (Mobility Management Entity):** Represented by a box icon, it is connected to the RNC and the GGSN.
- GGSN (Gateway GPRS support node):** Represented by a box icon, it is connected to the MME and the IP Network.
- NodeB (3G antenna):** Represented by a tower icon, it is connected to the BSC.
- NodeB-e (4G antenna):** Represented by a tower icon, it is connected to the BSC.



```
<?xml version="1.0"?>
<cdr>
  <id>b478fndh4dcdvjt</id>
  <start_time>2014-04-23 14:34:12</start_time>
  <start_epoch>1217529338</start_epoch>
  <start_uepoch>1217529338212616</start_uepoch>
  <answer_epoch>1217529338</answer_epoch>
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  <end_epoch>1217529377</end_epoch>
  <end_uepoch>1217529377795951</end_uepoch>
  <end_time>2014-04-23 14:35:14</end_time>
  <billsec>62</billsec>
  <success>1</success>
  <imsi>384845321548865</imsi>
  <type>call_out</type>
  <type_o>voice</type_o>
  <gen>2g</gen>
  <cgi>248487865145</cgi>
  <msisdn_rec>284872535485</msisdn_rec>
  <default_area_code>24807</default_area_code>
  <duration>00:01:02</duration>
  <record_type>125425</record_type>
  <accountcode>1160</accountcode>
  <user_context>default</user_context>
</cdr>
```

MNO Data Availability Checklist

MNO Data Availability Check List

The following list is used to identify the data available from MNOs and to propose the format and content of the data received from MNOs. This checklist should be filled by the MNO that is providing the source dataset.

NB! Make sure the check-list is read through and understood before filling!

Option	Explanation
+	Yes
-	No
?	Don't know
Text	Short free text answer

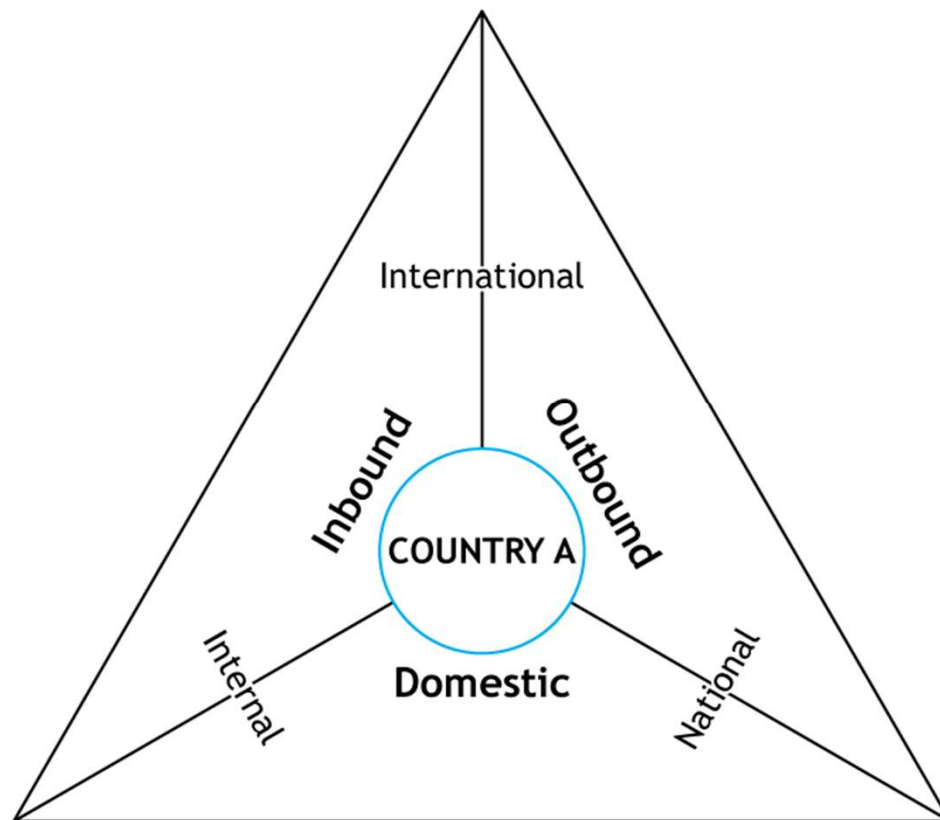
Domestic Records

Domestic records stored, available and provided by MNO.

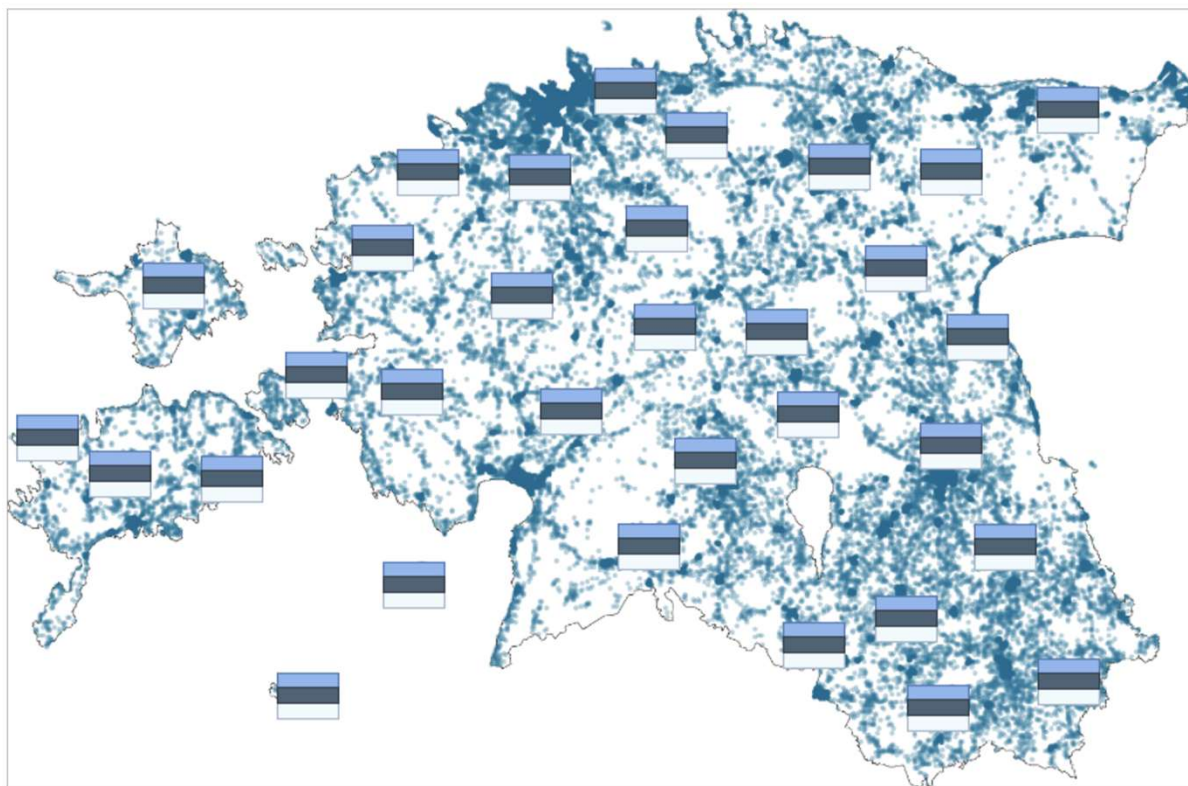
1. General	'+' or '-' or '?'	Where are these records stored (billing DB, etc.)?	How long are they stored (e.g. 1 month, 6 months, 2 years)?
1.1 Is CDR data available?			
1.2 Is IPDR data available?			
1.3 Is Core Network (CN) signalling data available?			
1.4 Is Radio Access Network (RAN) signalling data available?			

2. Which SIM card types are included in the data?	How to fill?
2.1 Are all post-paid SIM cards of the MNO included?	'+' or '-' or '?'
2.2 Are all post-paid SIM cards of the virtual MNO's operating in the network included?	'+' or '-' or '?'
2.3 Are all pre-paid SIM cards of the MNO included?	'+' or '-' or '?'
2.4 Are all pre-paid SIM cards of the virtual MNO' operating in the network included?	'+' or '-' or '?'
2.5 Are all international travel SIM cards issued by the MNO and used internationally included?	'+' or '-' or '?'

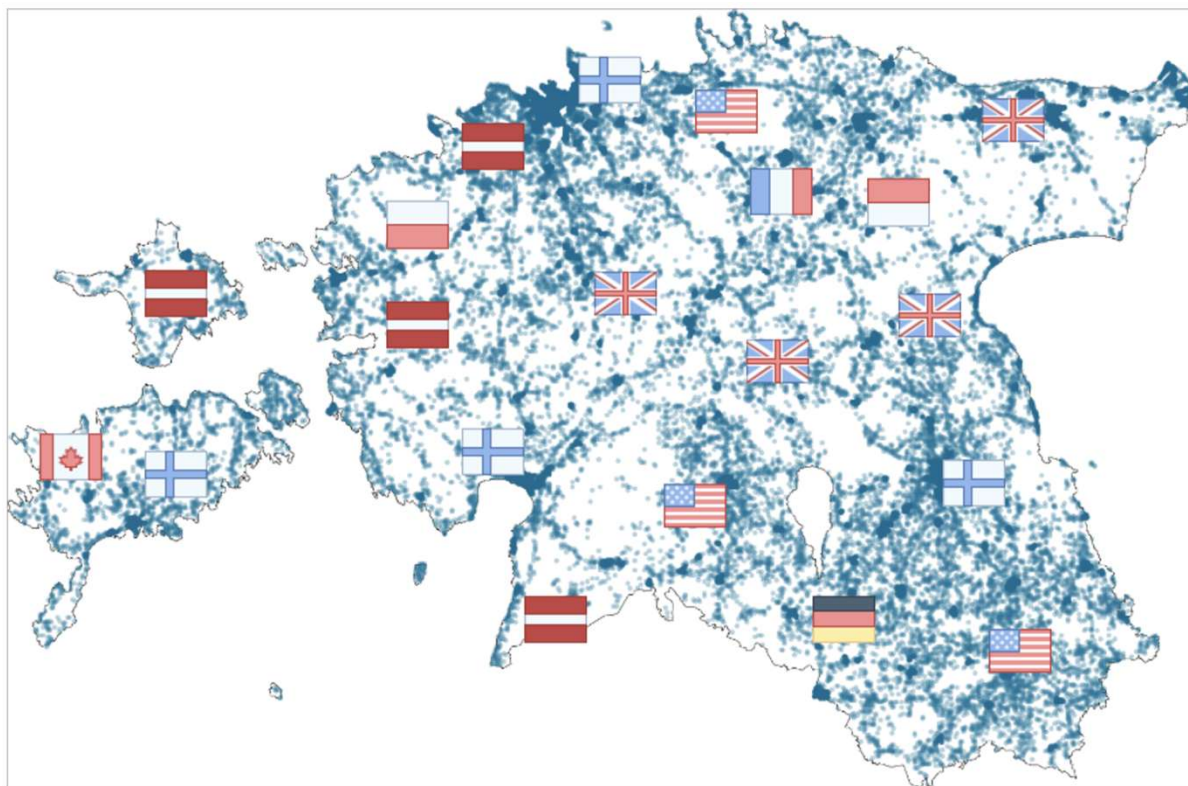
Domestic, Inbound, Outbound MPD



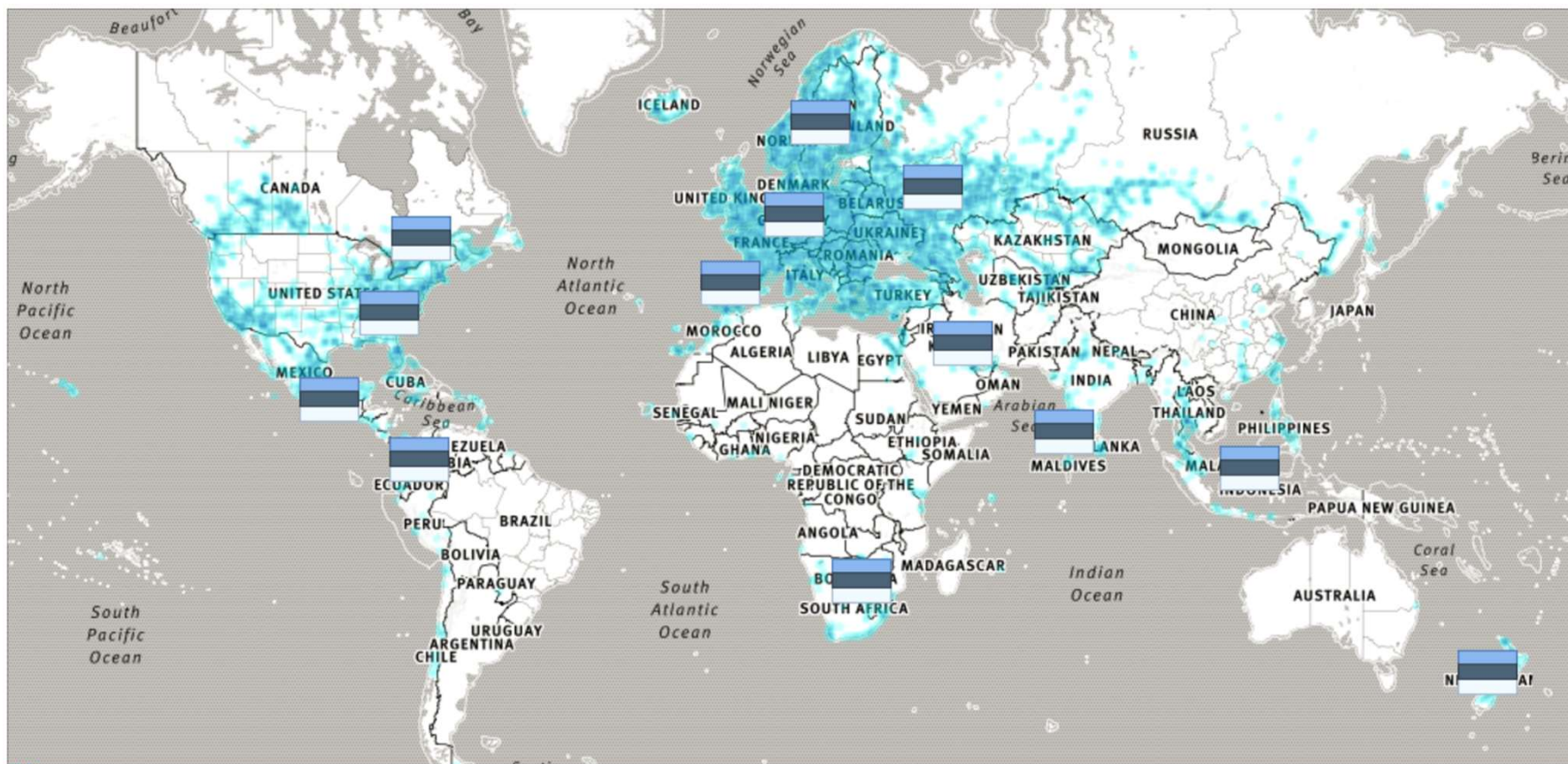
Domestic MPD



Inbound Roaming MPD (Inbound Tourism)



Outbound Roaming Data (Outbound Tourism)



Main Data Attributes

DOMESTIC	OUTBOUND	INBOUND
● ID	● ID	● ID ID for a person
● TIMESTAMP	● TIMESTAMP	● TIMESTAMP
● CELL_ID	● COUNTRY	● CELL_ID
● <i>(Home country)*</i>	● <i>CELL_ID**</i>	● COUNTRY (MCC from IMSI) Location attributes

* - if registration is required, then for pre-paid cards obtained by foreigners can be moved to inbound

** - in some cases, MNOs also store the CELL_ID of the event in a foreign country

Collection: Subscriber Identifier

Four possibilities of ID:

- International Mobile Subscriber Identity maps to Mobile Country Code
 - IMSI => MCC
- Mobile Station International Subscriber Directory Number maps to Country Code
 - MSISDN => CC
- International Mobile Equipment Identity maps to Type Allocation Code
 - IMEI => TAC
- MNO customer ID

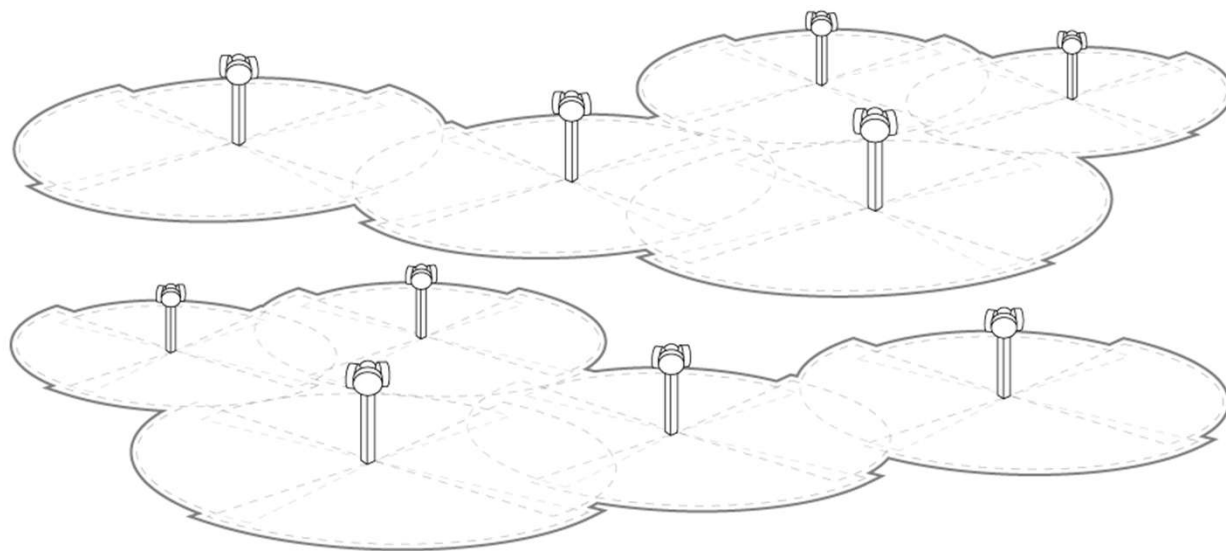
IMSI is preferred ID as raw data

It has to be hashed into **pseudonymous ID**

Collection: Network Antennae Data (Cell Data)

- Network antennae database includes:

- CELL_ID
- Type (2G, 3G, LTE)
- Location coordinates
- Coverage area (polygon)
- Height of the cell
- Power (db)
- Azimuth
- Angle



Collection: Additional Attributes (Optional)



Socio-demographic*:

- Age
- Gender
- Preferred language
- Address



IMEI

- Phone type
- IoT device



Contract type*

- pre-paid/post-paid
- private/commercial
- Travel SIM
- IoT device



Spending*

- Average Revenue Per User (ARPU)



Network*

(B-numbers)

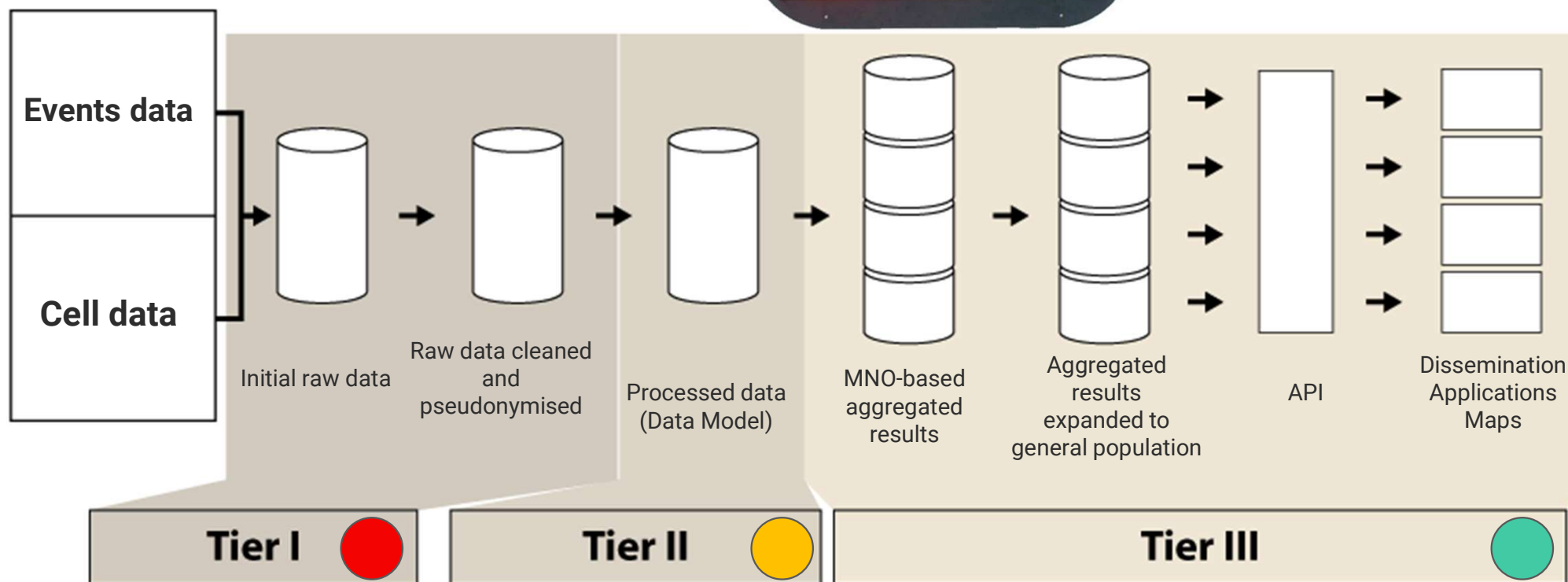
** May not be available or shared*

Privacy-by-Design in the Data Pipeline

3 tiers for MPD data in processing pipeline

- Tier I data: initial, raw, and not aggregated data containing:
 - personally identifiable information
 - business confidential information
- Tier II data: processing stage data:
 - no direct personally identifiable information
 - some business confidential information
 - data from different operators ready to be merged
- Tier III data: aggregated indicators that can be publicly shared:
 - no personally identifiable information
 - no business confidential information

3 tiers for MPD data



Structures of Data Pipelines

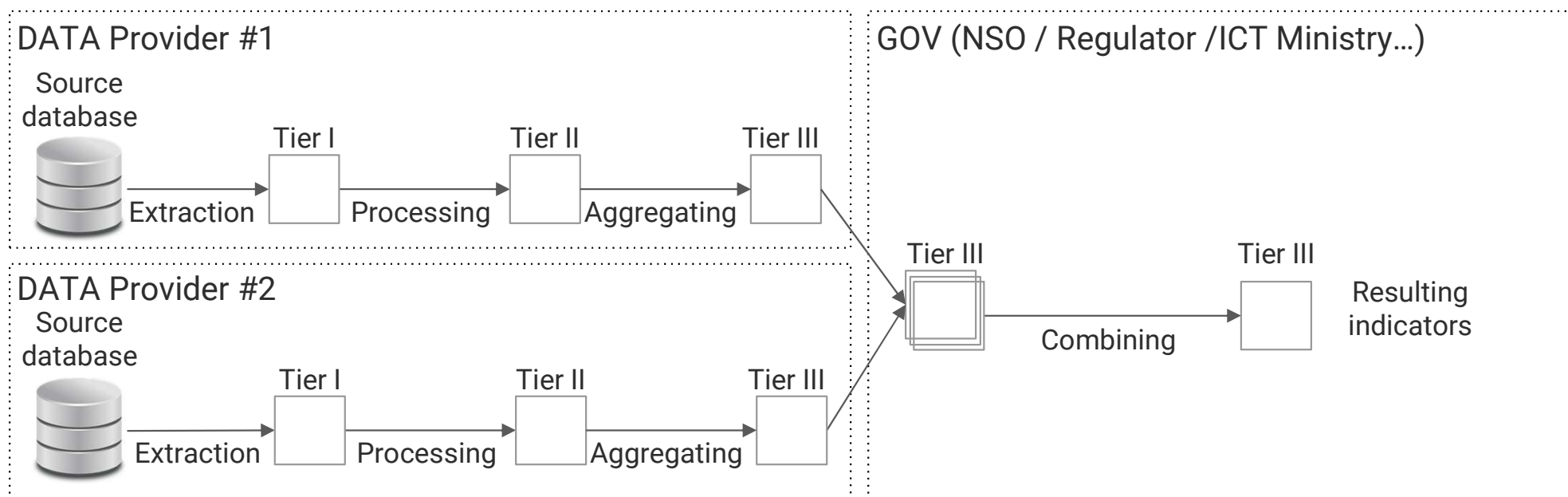
Data Processing Models

There are two main models based on where data processing is happening and who's responsible for the processing

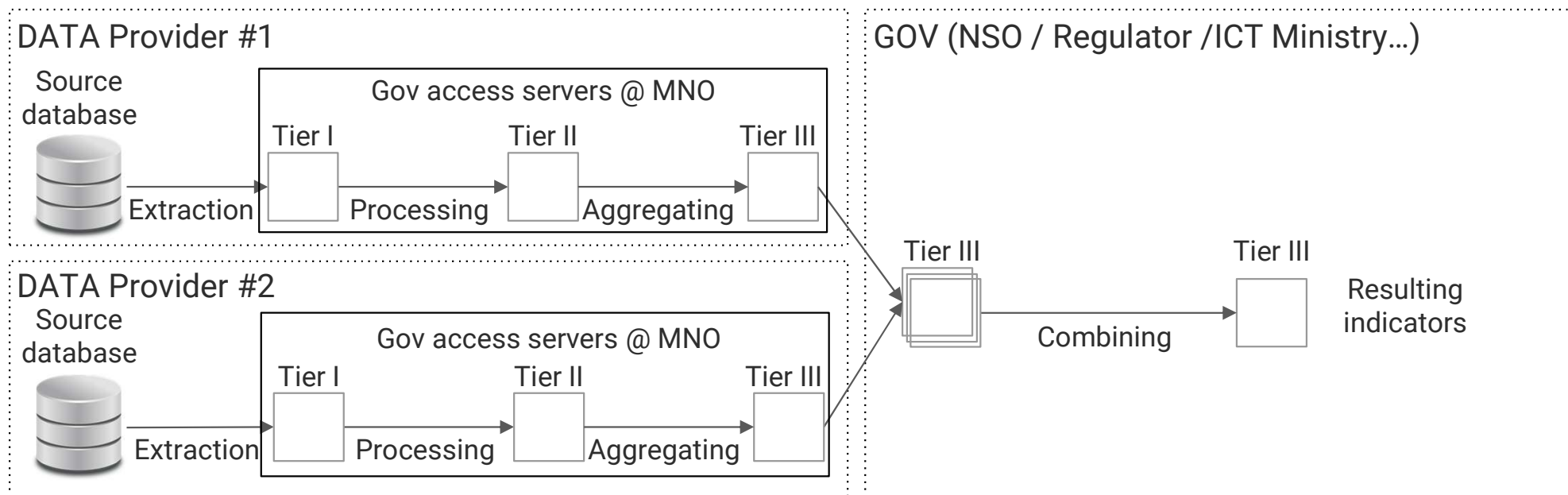
- Aggregation at MNO: MNOs mainly responsible for data processing (decentralized).
 - There is a submodel of this where MNO is responsible for technical infrastructure and government body or technical service provider is responsible for processing the data
- Aggregation at Government: Government stakeholders/non-operators mainly responsible for data processing (centralized)

Implications on the distribution of the costs & needed measures to ensure security, privacy and data protection

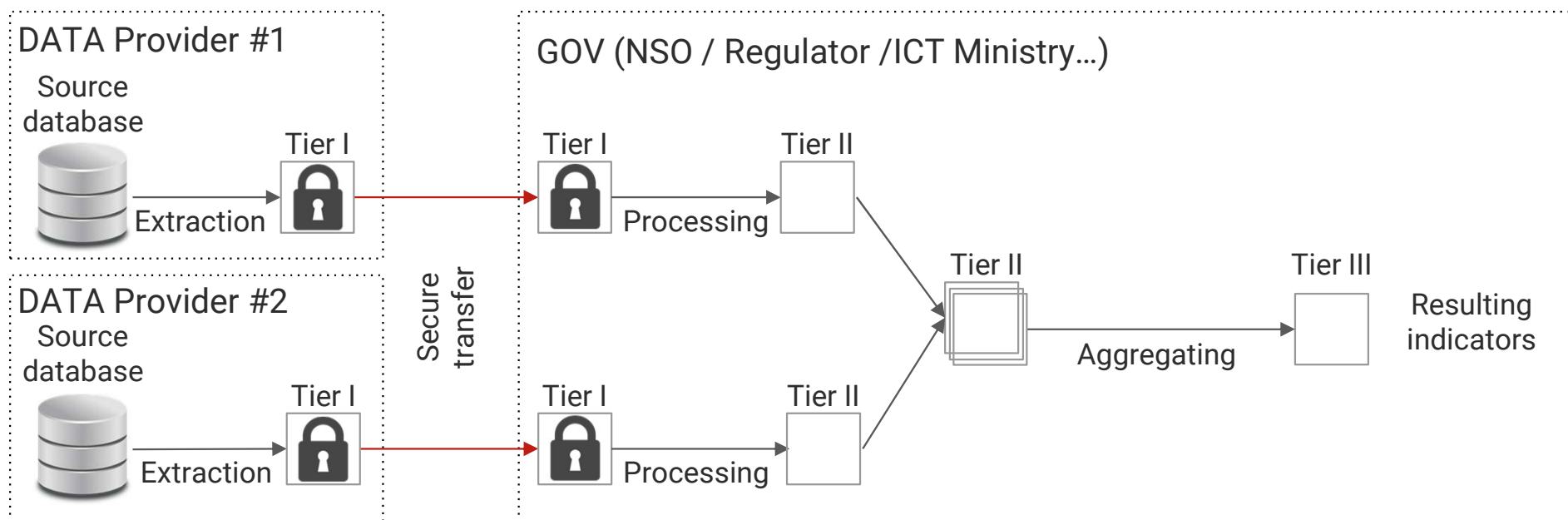
Aggregation at MNO by MNO



Aggregation at MNO by Government



Aggregation at Government



Aggregating at MNO by MNO

Advantages

- Higher data security / Lower risk of (accidental) disclosure
- Higher confidence for MNO

Disadvantages

- Higher cost for MNO
- Lower control/confidence for statistical authority
- Lower granularity in the data for statistical analysis

Aggregating at MNO by Government

Advantages

- Higher control/confidence for statistical authority (trust, transparency, accuracy)
- Higher data security

Disadvantages

- Very strong partnerships between stakeholders are needed
- Technical capacity needs to be built in each MNO with usually different server setups

Aggregating at Government

Advantages

- Higher control/confidence for statistical authority (trust, transparency, accuracy)
- Potential less disruption to statistical production when operators enter and exit the market

Disadvantages

- Statistical authority need to invest in IT infrastructure
- Higher privacy disclosure risks
- Legal basis may need to be set up

Advantages and disadvantages between models

Responsibility for most of the data processing	MNO	Gov / Third Party	Gov / Third Party
Granularity of data available for computing statistical indicators for Government	Low	High	High
Perception of risks to privacy and business confidentiality arising from use of data for official statistics	Low	Medium	High
Range of potential statistical indicators that can be derived	Low	High	High
Ease of verifiability and statistical quality control	Low	High	High
Impact of the entry and exit of new operators in the market on the production of official statistics	High	High	Low

GDF GLOBAL DATA FACILITY

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Thank you!



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